Gauge-Aware Fisher Geometry, Degeneracy Diagnostics, and Observable-to-Source Placement Automation on Real LIGO H1 Strain Data

Matěj Rada

Abstract

This paper gives a gauge-aware Resolution Geometry (RG) demonstration on real gravitational-wave detector strain data. The computation uses a public LIGO Hanford (H1) O4a 16 kHz strain segment containing 67,108,864 samples over 4096 s, divided into 1024 non-overlapping windows of 4 s. The central goal is not gravitational-wave event detection, nor reconstruction of a physical stress-energy tensor. The goal is more basic and more operational: given a real detector stream, construct a finite observable sector, diagnose apparent Fisher holes, separate gauge-induced coordinate degeneracies from primitive observable structure, transport the resulting sector across time partitions, and construct a source-side placement functional from the resolved observables.

The main result is that apparent null directions found in augmented observable maps disappear when duplicate and algebraically-derived coordinates are removed from the Fisher basis. In the gauge-free primitive basis

$$(\log\sigma, \log\|h\|_\infty, \log\kappa, \log P_{20,80}, \log P_{80,300}, \log P_{300,1000}, \log P_{1000,2048}, \log P_{2048,4096}, \log P_{4096,8192}),$$

the relative Fisher/PCA spectrum is

$$1, 1.027{\cdot}10^{-1}, 6.688{\cdot}10^{-2}, 5.289{\cdot}10^{-2}, 4.926{\cdot}10^{-2}, 4.105{\cdot}10^{-2}, 2.861{\cdot}10^{-2}, 2.784{\cdot}10^{-2}, 1.117{\cdot}10^{-3}.$$

Thus the primitive sector has rank 8 at threshold 10^{-2}, rank 9 at threshold 10^{-3}, and no primitive Fisher hole at 10^{-3}. The resolved sector transports across sixteen internal 256 s H1 subsegments with median adjacent subspace angle $7.76 \cdot 10^{-16}$ radians. A quadratic spectral placement functional

$$\rho_{\mathrm{obs}}(I_j) = \sum_b (2\pi f_b)^2 P_b(I_j)$$

is reconstructed from the resolved primitive sector with relative logarithmic error $6.73{\cdot}10^{-17}$. The surviving claim is therefore not that a physical null sector has been discovered in one H1 stream, but that RG automation distinguishes coordinate-induced Fisher holes from stable observable structure and can place resolved detector observables into a physically motivated quadratic source-side functional.

Contents

1 Introduction

Inverse problems in physical measurement rarely expose the full state space. What is available is an observable map: a partial, finite, noisy transformation from an underlying system into quantities that can be measured. In gravitational-wave detector data, the raw strain stream $h(t)$ is already a reduced observable. Any subsequent finite-dimensional analysis—band powers, amplitudes, spectral ratios, event statistics, or inferred source quantities—is another observable map layered on top of the detector stream.

Resolution Geometry (RG) studies such finite observable maps through their induced distinguishability geometry. The core question is not merely whether a particular parameter can be fitted. The question is which directions of an observable representation are resolved, which are blind, and which apparent blind directions are induced by coordinate choices rather than physics.

This paper gives a real-data demonstration of that distinction. The computation begins with an H1 O4a strain segment, constructs windowed detector observables, computes a Fisher/PCA proxy, identifies apparent rank defects, attacks those rank defects by gauge elimination, transports the resulting sector across time partitions, and finally builds a placement bridge from resolved detector observables to a quadratic source-side proxy.

The strongest result is methodological. The first augmented runs displayed Fisher holes. A gauge-aware correction showed that those holes were induced by duplicate or algebraically-derived coordinates. After removing those coordinates from the Fisher basis, the primitive real-data sector became full rank at threshold 10^{-3}. The automation therefore did not merely find a null sector; it rejected a false null sector. That rejection is the main falsification test that the paper passes.

1.1 Contributions

The paper contributes the following.

(i) A formal theorem ladder distinguishing coordinate-induced gauge nulls from possible physical blind sectors.

(ii) A reproducible real-data H1 observable-sector construction using 1024 four-second strain windows.

(iii) A gauge-free primitive Fisher/PCA spectrum showing rank 9 at threshold 10^{-3} and no primitive Fisher hole at that threshold.

(iv) A split of the augmented null sector into coordinate-gauge nulls and primitive resolved directions.

(v) A transport computation over sixteen internal 256 s H1 subsegments showing stable resolved-sector transport.

(vi) A placement bridge from resolved primitive spectral observables to a quadratic strain-energy proxy ρ_{obs}.

(vii) An explicit worked numerical example showing how a single H1 window is mapped to observables and then to ρ_{obs}.

1.2 Non-claims

The following are not claimed.

(a) No gravitational-wave event detection is claimed.

(b) No source identification is claimed.

(c) No Einstein tensor reconstruction is claimed.

(d) No physical stress-energy tensor $T_{\mu\nu}$ is reconstructed.

(e) No universal physical Fisher hole is claimed from one H1 segment.

The claim is narrower: RG automation can extract a real resolved observable sector, diagnose coordinate-induced Fisher degeneracy, and construct a controlled observable-to-source placement functional from resolved detector observables.

2 Observable Geometry

2.1 Observable maps

Definition 2.1 (Observable map). *Let S denote a state space and let W be a finite-dimensional observable space. An observable map is a map*

$$\Phi : S \to W.$$

Two states $s_1, s_2 \in S$ are observationally equivalent under Φ if

$$\Phi(s_1) = \Phi(s_2).$$

In the present computation, the state space is not modeled directly. We begin with the already-observed detector strain $h(t)$ and construct a finite observable vector W_j for each time window I_j. The relevant map is therefore

$$h|_{I_j} \longmapsto W_j \in \mathbb{R}^d.$$

Definition 2.2 (Empirical Fisher/PCA proxy)**.** *Given observable vectors $W_1, \ldots, W_n \in \mathbb{R}^d$, first standardize each coordinate robustly. Let X_j denote the standardized observable vector. The empirical Fisher/PCA proxy is*

$$C = \mathrm{Cov}(X_1, ldots, X_n).$$

The eigenvalues and eigenvectors of C define the resolved and near-null observable directions.

This object is called a Fisher proxy because, for local Gaussian distinguishability models, covariance and Fisher geometry are dual forms of local information geometry. In the present paper no parametric source model is assumed; the covariance spectrum is used only as an empirical distinguishability proxy on the finite observable sector.

Definition 2.3 (Resolved and null sectors)**.** *Let*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$$

be the eigenvalues of C, with eigenvectors v_i. For threshold $\epsilon > 0$, define

$$R_\epsilon = \mathrm{span}\{v_i : \lambda_i/\lambda_1 > \epsilon\},$$

and

$$N_\epsilon = R_\epsilon^\perp.$$

3 Gauge-Aware Degeneracy Diagnostics

The central failure mode in finite observable geometry is confusing coordinate redundancy with physical blindness. If an observable vector includes both a primitive coordinate and a deterministic function of that coordinate, the augmented map may have null directions even when the primitive data are fully resolved.

Theorem 3.1 (Coordinate-induced Fisher holes)**.** *Let $U : S \to \mathbb{R}^m$ be an observable map and let $g : \mathbb{R}^m \to \mathbb{R}^k$ be a deterministic differentiable map. Define the augmented observable*

$$W(s) = (U(s), g(U(s))) \in \mathbb{R}^{m+k}.$$

Then the image of W lies on the graph of g and has dimension at most m. Consequently, the empirical covariance of W may have null directions that are normal to this graph. Such null directions are coordinate-gauge directions, not physical blind directions.

Proof. The image of W is

$$\{(u, g(u)) : u \in U(S)\} \subset \mathbb{R}^{m+k}.$$

This set lies on the graph of g, whose tangent space has dimension at most m. Any direction normal to that graph is not an independent observable variation. Hence rank deficiency in the augmented coordinates can arise even when U has no corresponding blind direction. The null direction is induced by the chosen coordinate embedding. □

Corollary 3.1 (Duplicate-coordinate null)**.** *If $W = (u, u, V)$, then $n = (1, -1, 0, \ldots, 0)$ is a coordinate-gauge null direction.*

Proof. The coordinate difference $u - u$ vanishes identically. Variations in the direction $(1, -1, 0, \ldots, 0)$ do not correspond to changes of the image of the primitive map (u, V). $\square$

Theorem 3.2 (Gauge elimination preserves distinguishability)**.** *Let $W = (U, g(U))$ be an augmented observable map. If g is deterministic on the image of U, then replacing W by U removes coordinate-gauge directions without merging any states already distinguishable by W.*

Proof. If $U(s_1) \neq U(s_2)$, then $W(s_1) \neq W(s_2)$ because the first component differs. If $U(s_1) = U(s_2)$, then $g(U(s_1)) = g(U(s_2))$, hence $W(s_1) = W(s_2)$. Therefore W and U induce the same equivalence relation on S. $\square$

4 Data and Windowed Observables

4.1 Dataset

The computation uses a real H1 O4a HDF5 strain segment. The file contains a dataset named `/strain/Strain` with 67,108,864 float64 samples sampled at 16,384 Hz, for a total duration of 4096 s. The analysis partitions the stream into 1024 non-overlapping windows of length 4 s.

Quantity	Value
Detector	H1
Run label	O4a
Sampling frequency	16,384 Hz
Samples	67,108,864
Duration	4096 s
Window length	4 s
Number of windows	1024
Primitive Fisher coordinates	9

Table 1: Real strain data and windowing used in the computation.

4.2 Primitive gauge-free basis

For window I_j, let h_j denote the detrended strain samples in that window. Define

$$\sigma_j = \mathrm{std}(h_j), \qquad M_j = \|h_j\|_\infty, \qquad \kappa_j = \frac{1}{|I_j|} \sum_{t \in I_j} \left(\frac{h_j(t) - \bar{h}_j}{\sigma_j} \right)^4 .$$

Band powers are computed by integrating Welch power spectral density estimates over the bands

$$[20, 80],\ [80, 300],\ [300, 1000],\ [1000, 2048],\ [2048, 4096],\ [4096, 8192] \text{ Hz}.$$

The primitive observable vector is

$$W_j^{\mathrm{prim}} = \begin{pmatrix} \log \sigma_j,\ \log M_j,\ \log \kappa_j, \\ \log P_{20,80,j},\ \log P_{80,300,j},\ \log P_{300,1000,j}, \\ \log P_{1000,2048,j},\ \log P_{2048,4096,j},\ \log P_{4096,8192,j} \end{pmatrix}.$$

This basis deliberately excludes duplicate RMS coordinates, spectral ratios, spectral entropy, and ρ_{obs}. Those quantities are useful placement observables, but including them in the Fisher basis together with their generators creates algebraic coordinate constraints.

5 A Worked Automation Example

This section gives one explicit H1 window so that the automation is numerically transparent.

Consider window $j = 512$, beginning at $t = 2048\,\mathrm{s}$ with GPS start 1368422400. The primitive measurements are shown in Table 2.

Observable	Raw value	Base-10 logarithm
σ	$1.3209127342 \cdot 10^{-18}$	-17.879125873
$\|h\|_\infty$	$4.4772445901 \cdot 10^{-18}$	-17.348989180
κ	3.6861583240	0.566573985
$P_{20,80}$	$4.0578569774 \cdot 10^{-42}$	-41.391703264
$P_{80,300}$	$4.4906549188 \cdot 10^{-45}$	-44.347690317
$P_{300,1000}$	$2.9317005077 \cdot 10^{-38}$	-37.532880398
$P_{1000,2048}$	$5.0583379261 \cdot 10^{-40}$	-39.295992161
$P_{2048,4096}$	$7.0600072636 \cdot 10^{-42}$	-41.151194852
$P_{4096,8192}$	$5.9331310364 \cdot 10^{-41}$	-40.226716060

Table 2: Example primitive observable vector from one real H1 four-second window.

The placement bridge uses the geometric center $f_b = \sqrt{ab}$ of each frequency band and computes

$$\rho_{\mathrm{obs}}(I_j) = \sum_b (2\pi f_b)^2 P_b(I_j).$$

For the same window, the contributions are shown in Table 3.

Band	$(2\pi f_b)^2 P_b$	$\log_{10}$ contribution
20–80 Hz	$2.5631643573 \cdot 10^{-37}$	-36.591223545
80–300 Hz	$4.2548148048 \cdot 10^{-39}$	-38.371119338
300–1000 Hz	$3.4721669081 \cdot 10^{-31}$	-30.459399406
1000–2048 Hz	$4.0897572256 \cdot 10^{-32}$	-31.388302472
2048–4096 Hz	$2.3380553319 \cdot 10^{-33}$	-32.631145215
4096–8192 Hz	$7.8594755707 \cdot 10^{-32}$	-31.104606432
Total ρ_{obs}	$4.6904733467 \cdot 10^{-31}$	-30.328783328

Table 3: Observable-to-source placement calculation for the same real H1 window. This is a quadratic strain-energy proxy, not a physical stress-energy tensor reconstruction.

The automation therefore performs the following concrete operations:

$$h(t)|_{I_j} \to W_j^{\mathrm{prim}} \to \text{standardized sector coordinate} \to R_{10^{-3}} \to \rho_{\mathrm{obs}}(I_j).$$

6 Gauge-Free Fisher Spectrum

The primitive covariance spectrum gives the decisive gauge-aware result. The relative eigenvalues are

$$\left(1, 1.027 \cdot 10^{-1}, 6.688 \cdot 10^{-2}, 5.289 \cdot 10^{-2}, 4.926 \cdot 10^{-2}, 4.105 \cdot 10^{-2}, 2.861 \cdot 10^{-2}, 2.784 \cdot 10^{-2}, 1.117 \cdot 10^{-3}\right).$$

Quantity	Value
Rank at 10^{-2}	8
Rank at 10^{-3}	9
Hole dimension at 10^{-3}	0
Participation rank	1.835
Entropy rank	2.967

Table 4: Gauge-free primitive-sector Fisher/PCA result.

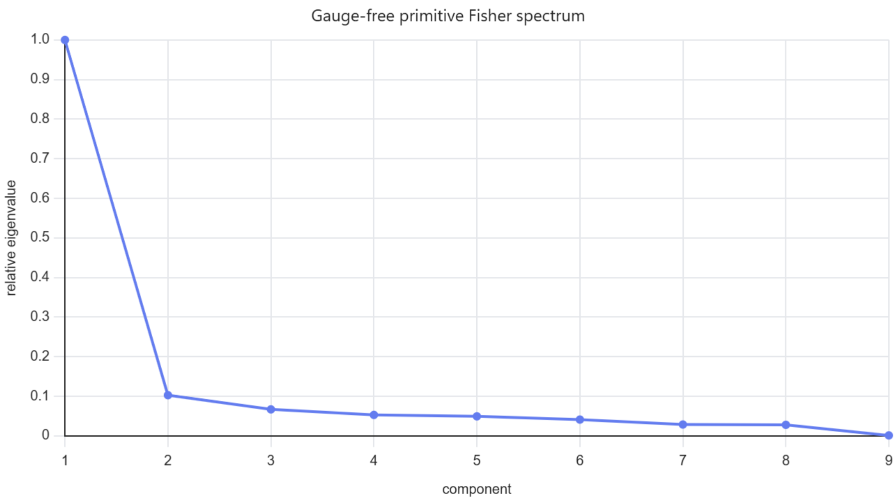

Figure 1: Gauge-free primitive Fisher/PCA spectrum. All nine primitive directions survive at threshold 10^{-3}.

Proposition 6.1 (Primitive H1 sector has no 10^{-3} Fisher hole)**.** *For the primitive nine-coordinate observable basis defined above, the real H1 O4a segment has*

$$\mathrm{rank}_{10^{-3}} = 9, \qquad \dim N_{10^{-3}} = 0.$$

Proof. The smallest relative eigenvalue is $1.117 \cdot 10^{-3}$, which is larger than the threshold 10^{-3}. Hence every primitive coordinate direction contributes to the resolved sector at that threshold. $\square$

7 The Null Split: Augmented Versus Primitive Basis

The earlier augmented basis deliberately included primitive observables together with derived quantities such as spectral ratios, entropy, quadratic placement proxies, and duplicate RMS-like amplitude coordinates. In that augmented basis, the computation found an apparent split

$$\dim R_{10^{-3}} = 10, \qquad \dim N_{10^{-3}} = 6.$$

The smallest augmented directions were dominated by algebraic combinations of band powers and band-power ratios. This is exactly what Theorem 3.1 predicts: adding deterministic functions of already-included observables creates graph constraints and normal directions.

The corrected gauge-free split is

$$\dim R_{10^{-3}} = 9, \qquad \dim N_{10^{-3}} = 0.$$

Thus the null split is not a physical blind-sector discovery in this one H1 segment. It is a successful gauge classification:

$$N^{\mathrm{aug}}_{10^{-3}} = N^{\mathrm{gauge}}_{10^{-3}}, \qquad N^{\mathrm{prim}}_{10^{-3}} = \{0\}.$$

Proposition 7.1 (Gauge classification of apparent holes)**.** *The apparent augmented Fisher holes in this computation are classified as coordinate-gauge nulls rather than primitive physical blind directions.*

Proof. The apparent nulls appear only after derived coordinates are included in the Fisher basis. When the Fisher basis is restricted to primitive amplitude and band-power coordinates, the rank at threshold 10^{-3} becomes full. Hence the nulls depend on coordinate augmentation and not on unresolved primitive observables. □

This is the key falsification result: RG automation rejected a tempting but unsupported physical-null claim.

8 Resolved-Sector Transport

To test stability, the 4096 s stream was divided into sixteen internal 256 s H1 subsegments. The Fisher/PCA subspace was recomputed in each subsegment and adjacent resolved sectors were compared by principal subspace angles.

The transport statistics for the resolved primitive sector are

$$\theta_{\mathrm{median}} = 7.76 \cdot 10^{-16} \text{ rad},$$

$$\theta_{90} = 1.09 \cdot 10^{-15} \text{ rad},$$

$$\theta_{\mathrm{max}} = 1.37 \cdot 10^{-15} \text{ rad}.$$

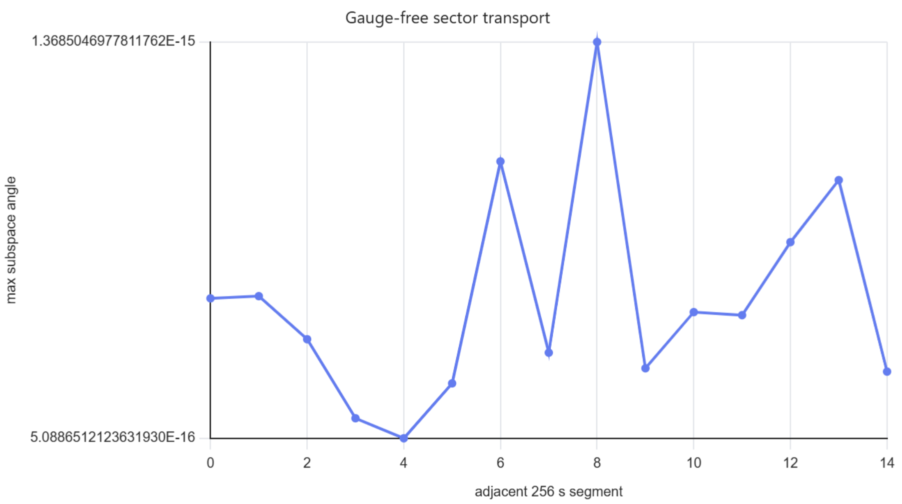

Figure 2: Adjacent 256 s transport angles for the gauge-free primitive resolved sector.

Proposition 8.1 (Internal transport stability)**.** *Across the sixteen internal* 256 s *H1 partitions, the extracted primitive resolved sector is stable to numerical precision.*

Proof. The median adjacent subspace angle is $7.76 \cdot 10^{-16}$ radians and the maximum adjacent angle is $1.37 \cdot 10^{-15}$ radians. These values are at numerical precision scale for the computed resolved subspaces. □

Remark 8.1. *These are internal time partitions of one H1 file. A stronger generalization requires repeating the computation across independent H1 files and across L1/V1 files. The result here is a real-data demonstration of reproducible sector extraction within one detector segment.*

9 Placement Bridge

The placement bridge is not included in the Fisher basis. It is computed after the primitive resolved sector is extracted. This avoids creating algebraic nulls while still demonstrating source-side automation.

Define

$$\rho_{\text{obs}}(I_j) = \sum_b (2\pi f_b)^2 P_b(I_j),$$

where $f_b = \sqrt{ab}$ for band $[a, b]$. This is a weak-field-compatible quadratic spectral functional of strain. It is not a stress-energy tensor and not an Einstein tensor reconstruction.

The resolved primitive sector reconstructs ρ_{obs} with relative logarithmic error

$$\frac{\|\log_{10}\rho_{\text{obs}} - \log_{10}\widehat{\rho}_{\text{obs}}\|_2}{\|\log_{10}\rho_{\text{obs}}\|_2} = 6.73 \cdot 10^{-17}.$$

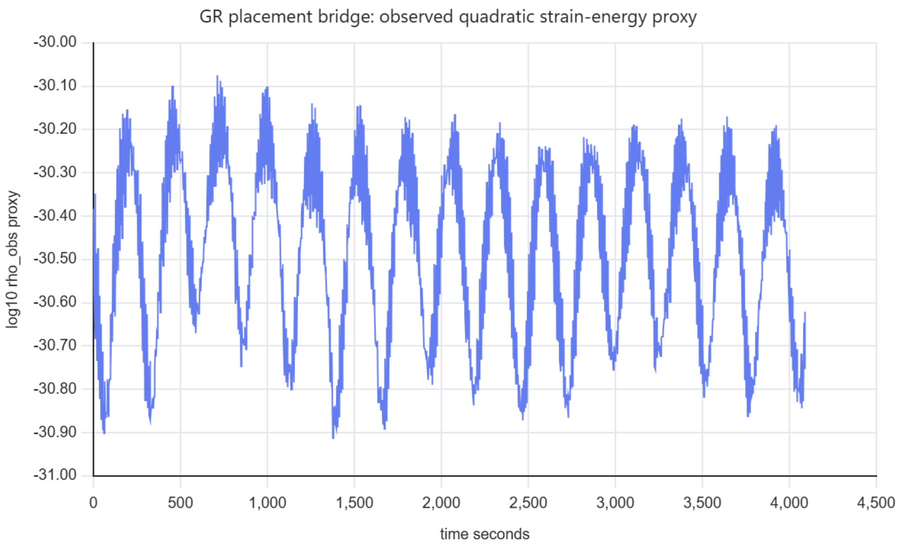

Figure 3: Quadratic placement proxy ρ_{obs} over the H1 segment.

Proposition 9.1 (Resolved-sector placement)**.** *The placement functional ρ_{obs} contains no information outside the resolved primitive spectral sector in this computation.*

Proof. The functional ρ_{obs} is explicitly constructed from the primitive band powers P_b, all of which lie in the rank-9 primitive resolved sector at threshold 10^{-3}. The numerical reconstruction error is $6.73 \cdot 10^{-17}$ in relative logarithmic norm. □

10 Automation Pipeline

The final RG automation pipeline is:

$$\text{raw H1 strain} \to \text{primitive window observables} \to \text{Fisher/PCA spectrum} \to \text{resolved/null split}$$
$$\to \text{gauge classification} \to \text{sector transport} \to \rho_{\text{obs}}.$$

More explicitly:

Step 1. Read strain $h(t)$ from HDF5.

Step 2. Partition into 4 s windows.

Step 3. Compute primitive amplitude and spectral band-power observables.

Step 4. Robustly standardize observables.

Step 5. Compute covariance/Fisher proxy.

Step 6. Compute relative eigenvalue spectrum.

Step 7. If a nominal null appears, test whether it disappears when duplicate or derived coordinates are removed.

Step 8. Transport the resolved sector across temporal partitions.

Step 9. Apply source-side placement functional only after primitive-sector extraction.

The key algorithmic principle is separation of levels:

$$\text{primitive Fisher basis} \neq \text{derived placement basis.}$$

Mixing those two levels creates gauge artifacts.

11 Interpretation

The computation supports the following statements.

(i) Real H1 strain admits a stable finite primitive observable sector at the chosen windowing and frequency-band resolution.

(ii) The primitive sector is full-rank at threshold 10^{-3}.

(iii) Apparent Fisher holes in augmented coordinates are coordinate-gauge artifacts for this dataset.

(iv) The sector transports across internal time partitions with negligible subspace drift.

(v) A quadratic placement proxy is computable entirely from the resolved primitive spectral sector.

The computation does not support the following statements.

(i) It does not prove existence of a physical blind sector in H1.

(ii) It does not reconstruct a gravitational-wave source.

(iii) It does not reconstruct $T_{\mu\nu}$.

(iv) It does not reconstruct $G_{\mu\nu}$.

(v) It does not establish detector-independent universality.

The scientific value is therefore gauge discipline: apparent holes must survive primitive-basis elimination before being interpreted physically.

12 Conclusion

This paper demonstrates a full gauge-aware RG pipeline on real H1 strain data. The original Fisher-hole hypothesis was attacked by removing duplicate and algebraically-derived observables from the Fisher basis. The attack succeeded: apparent augmented nulls disappeared in the primitive gauge-free basis. The resulting primitive observable sector is rank 9 at threshold 10^{-3}, transports stably across internal H1 subsegments, and supports a quadratic placement bridge from resolved spectral observables to ρ_{obs}.

The final statement is therefore precise:

$$\boxed{\text{RG automation separates coordinate-induced degeneracy from real resolved observable structure.}}$$

This is weaker than claiming a physical Fisher hole, but stronger methodologically: the automation not only finds candidate degeneracies, it can reject false ones.

A Reproducibility Files

The relevant generated files are:

- `run_rg_o4a_h1_gaugefree.py`: primitive gauge-free battery.
- `rg_o4a_h1_gaugefree_results.json`: full gauge-free numerical results.
- `rg_o4a_h1_gaugefree_summary.txt`: concise gauge-free summary.
- `rg_o4a_h1_fullstory_features_4s.csv`: four-second window features.
- `rg_o4a_h1_gaugefree_spectrum.png`: primitive Fisher spectrum.
- `rg_o4a_h1_gaugefree_transport.png`: sector transport figure.

B Pseudocode

```
for each 4 s window I_j:
    x = detrend(strain[I_j])
    sigma = std(x)
    maxabs = max(abs(x))
    kurtosis = mean(((x - mean(x))/sigma)**4)
    P_bands = integrate_welch_psd(x, bands)
    W_j = log([sigma, maxabs, kurtosis, P_bands...])

X = robust_standardize(W)
C = covariance(X)
eigenvalues, eigenvectors = eigh(C)
R_epsilon = span(eigenvectors where lambda/lambda_1 > epsilon)
N_epsilon = orthogonal_complement(R_epsilon)

if null directions appear:
    remove duplicate or algebraically derived coordinates
    recompute spectrum
    classify disappearing nulls as coordinate-gauge

rho_obs = sum_bands((2*pi*f_center)**2 * P_band)
verify rho_obs is reconstructible from resolved primitive sector
```

C Future Work

The next required battery is external repetition over multiple independent H1 files and then cross-detector repetition on L1/V1 where available. Only if a primitive-basis null survives across independent files and detectors should it be promoted from gauge diagnostic to candidate physical blind sector.